

Faculty of Engineering and Technology
B.Tech (CS&E)
MOOC2 – Core Python

TASK - 1

1. Write a program to create a simple calculator to perform all the arithmetic operators in python.

Addition - a+b

Subtraction - a-b

Multiplication - a*b

Division -a/b

Floor division – a//b

Modulo – a%b

Power — a**b

NOTE: Use assignment operators to assign the values to variables like a+=b.

2. Use Comparison and logical operators to perform the below conditions

- Ask the user for the bill amount and the number of friends to split the bill with.
- If the number of friends is 0 or more than the bill amount, tell the user to start over.
- If each person's share of the bill is more than 50 Rs, ask the user to enter a smaller number of friend
- If the above conditions are met, calculate the total bill including 20% tax, then divide it by the number of friends.
- Round off the final amount per person to two decimal points using the round() function

3. Write a program to show the use of bitwise operators

Bitwise AND – &

Bitwise OR – |

Bitwise NOT – ~

Bitwise XOR – ^

Bitwise right shift – >>

Bitwise left shift – <<

Special operators

1.Identity operator (is and is not)

2.Membership operator (in and not in)

SAMPLE EXAMPLE

```
fruits = ["apple", "banana", "cherry", "date"]
```

```
# Check if 'banana' is in the list of fruits
```

```
if "banana" in fruits:
```

```
    print("Banana is in the list!")
```

```
else:
```

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```
print("Banana is not in the list.")
```

4. Use of ternary operator in python to find whether the given number is odd or even.

Syntax:

value_if_true if condition else value_if_false

Example :

```
a = 10
```

```
b = 20
```

```
# Using the ternary operator to assign the greater value to 'max_value'
```

```
max_value = a if a > b else b
```

```
print("The greater value is:", max_value)
```

5. Write a program to create a Guessing Game

Note: This program allows the user to guess a number between 1 and 10, providing feedback until the correct number is guessed using a while loop.

6. Write a Python Program to print below mentioned patterns.

